Rosha Razmarafar
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Research Interests

- **Deep learning** for stochastic processes and dynamical systems modeling
- **AI-driven simulation** and scientific machine learning for modeling physical and complex systems
- **Generative modeling** (diffusion and score-based methods) for trajectory generation in complex systems
- **Reinforcement learning** for control and **decision-making** in stochastic and emergent environments
- **Autonomous** and **intelligent agents** in multi-agent and adaptive systems

Education

PhD in Computational Science and Engineering (Direct PhD) 2025 – Present
Koç University, Turkey
GPA: 3.74 / 4.0

B.Sc. in Computer Engineering 2020 – 2024
Maltepe University, Turkey
GPA: 3.69 / 4.0

Associate Degree in Computer Programming (Double Major) 2021 – 2024
Maltepe University, Turkey
GPA: 3.74 / 4.0

Honors & Awards

- 1st Rank, Faculty of Engineering — Maltepe University (2021–2024)
- 1st Place, MathWorks Minidrone Competition — 2024
- 4th Place, MathWorks Minidrone Competition — 2023

Publications & Work in Progress

Neural Score Models for Control of Stochastic Dynamical Systems (Active Ornstein–Uhlenbeck Processes) (In preparation)

Developing learning-based methods for modeling and controlling stochastic diffusion processes using neural approximations, with applications to high-dimensional dynamical systems.

Research Experience

Real-Time Air Pollution Monitoring via WSN Simulation

Designed and implemented an end-to-end real-time air pollution monitoring system based on a large-scale wireless sensor network (WSN) simulation, developed independently. The system models distributed sensor nodes and employs hierarchical clustering with the LEACH protocol for efficient data aggregation and transmission. Built a multi-layer architecture integrating simulation, data processing, and visualization, including a centralized database and web-based interface for real-time analysis and monitoring.

Centralized Cybersecurity Attack Platform (CCAP)

Designed and implemented a sandboxed platform for simulating cybersecurity attacks, including Remote Code Execution (RCE) and network-level behaviors. Integrated ASP.NET MVC with Windows Services and ARP-based scanning to model and analyze adversarial scenarios, enabling controlled evaluation of system vulnerabilities.

Teaching Experience

Teaching Assistant — PHYS 102 Spring 2025, Fall 2025, Spring 2026

Koç University

- Conducted problem-solving sessions in undergraduate physics
- Supported students and reinforced interdisciplinary links between physics and computation

Technical Expertise

Machine Learning & AI: PyTorch, deep learning models and algorithms, generative modeling (diffusion / score-based models), optimization methods

Reinforcement Learning: RL Algorithms; experience applying RL for control of complex and emergent physical systems

Scientific Computing & Simulation: Monte Carlo methods, stochastic processes, dynamical systems, soft matter and active matter simulations, OMNET++

Programming: Python, C, C++, C#, Julia, JAVA, MATLAB

Systems & Tools: Git, Linux, numerical modeling, distributed systems fundamentals

Additional: Experience refactoring research codebases (Julia → Python), implementing scalable experimental pipelines, and integrating simulation with learning-based models

Languages

- Persian — Native
- English — Advanced (TOEFL 101: Reading 28, Listening 22, Writing 23, Speaking 28)
- Turkish — Professional Working Proficiency
- German — Beginner–Intermediate

Research Collaborations

- Dr. Livio Nicola Carenza — Koç University `lcarenza@ku.edu.tr`